



Economics

Production and Market Structure

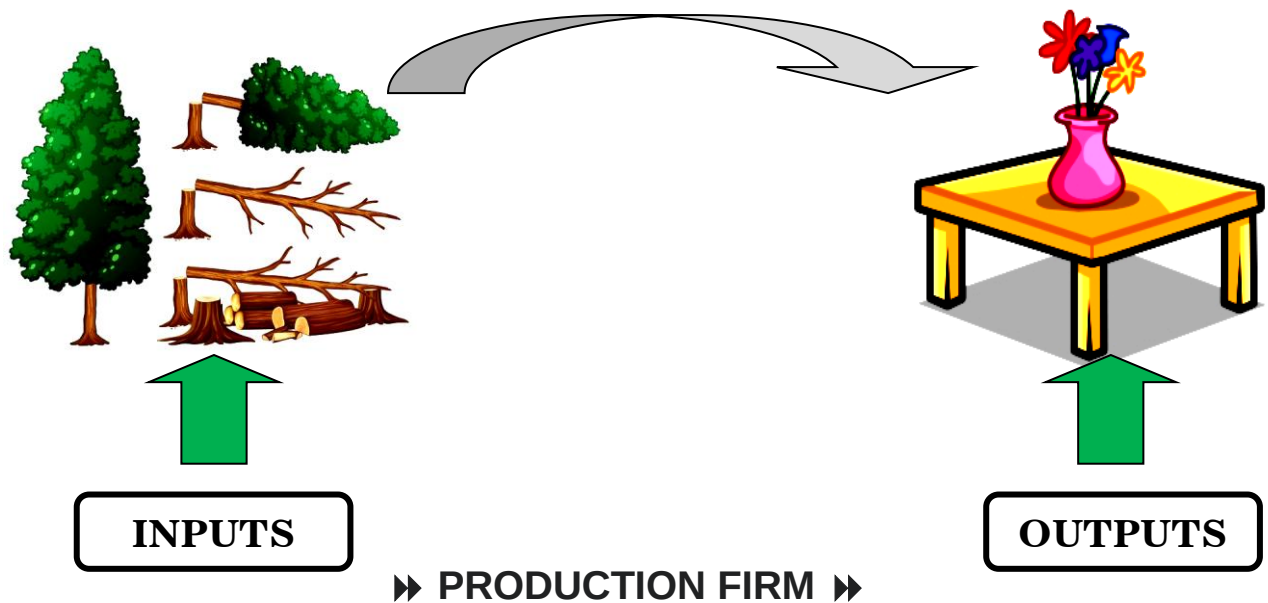
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GCE ADVANCED LEVEL ENGLISH MEDIUM

Production

Production is a process of combining various inputs in order to make something for consumption (output).

The process of converting inputs into outputs through a certain technological process is called production process.

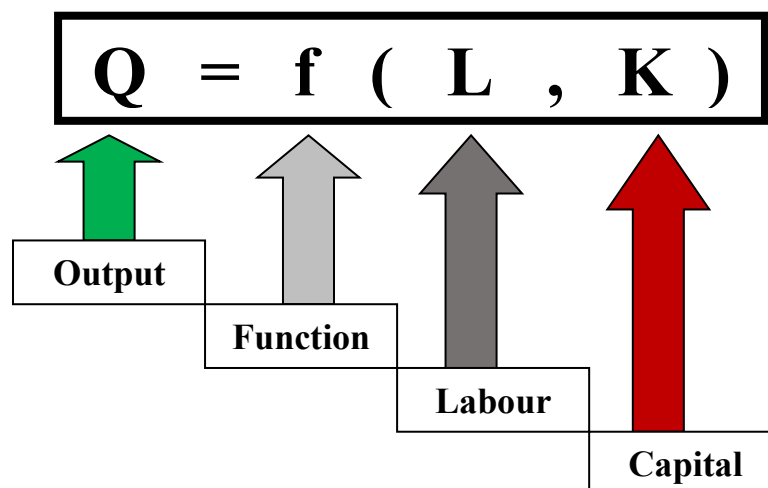


A unit which combines necessary resource in this way and a unit which organizes production is called a production firm.

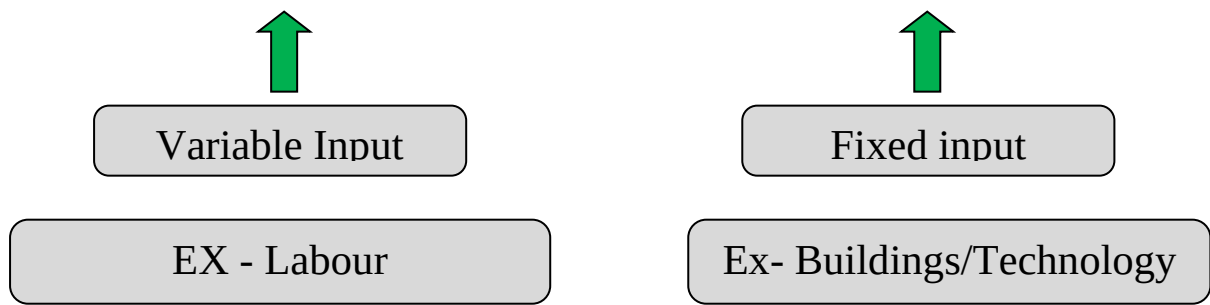
- Any production firm's main objective is to maximize profit.

» PRODUCTION FUNCTION »

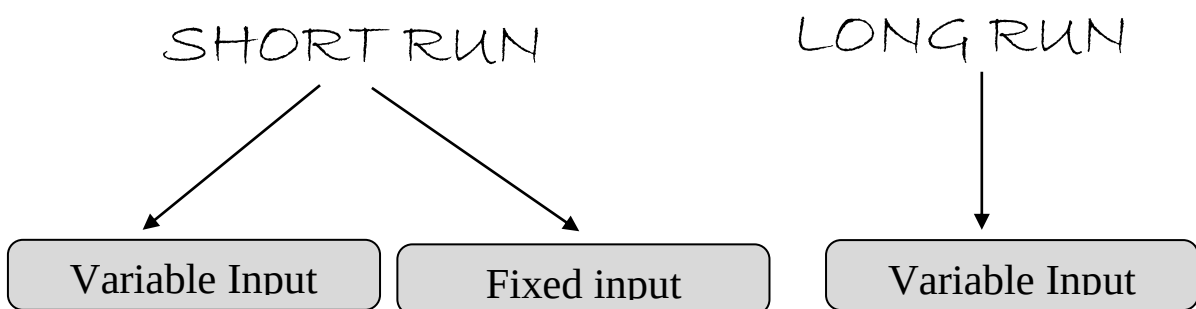
The technical relationship between inputs and outputs is called a production function.



some inputs **can be changed fast** and some inputs **cannot be changed** accordingly.



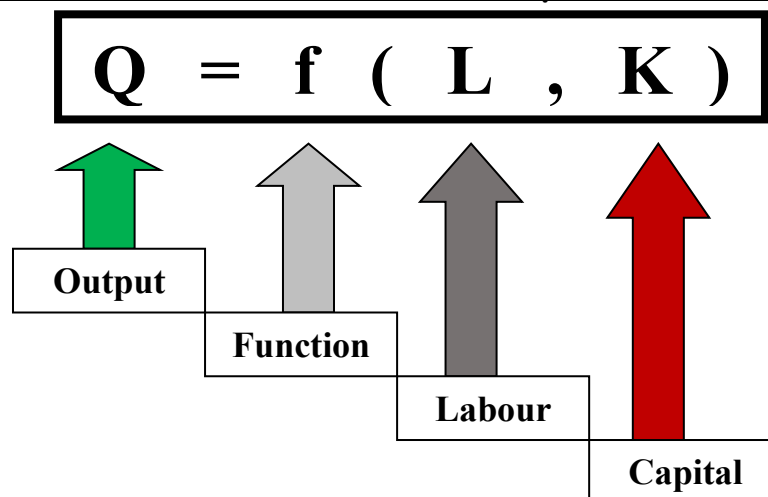
In this way two time periods relating to production are identified depending upon the time taken to change inputs.



» Short run production process »

Two factors called fixed input and variable inputs can be seen in the short run.

Production function of Short run production process



Under this we can identify 3 Main concepts.

» Total Product (TP)

The total output by mixing variable factors and fixed factors can be known as total production.

When engaged in production mixing variable factors with fixed factors, the output result is identified as total product.

» Average Product (AP)

Quantity produced by one worker from the total output can be known as average production.

$$AP = \frac{TP}{L}$$

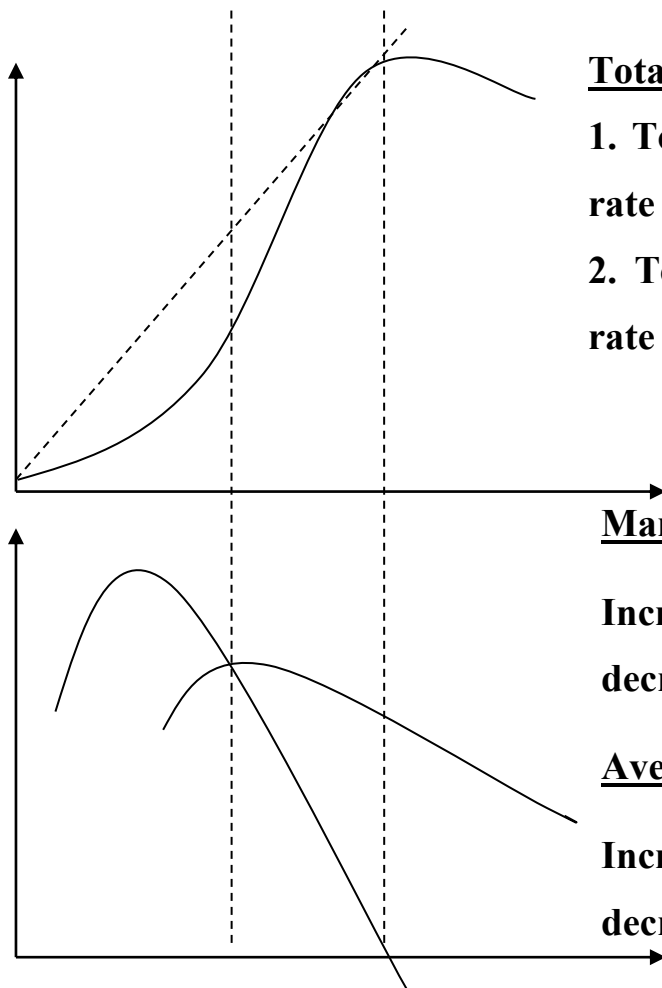
» Marginal Product (MP)

Marginal product of labor (MP) is the change in output that results from employing an added unit of labour

$$MP = \frac{\Delta TP}{\Delta L}$$

Complete the table below using the idea you got from the above concepts.

INPUTS		OUTPUT		
Labour Variable factors	Capital Fixed factors	Total product	Average product AP	Marginal product MP
0	10	0	-	
1	10	3		3
2	10	10		7
3	10	24		
4	10	36		
5	10	40		
6	10	42		
7	10	42		
8	10	40		



Total product

1. Total Product Increases at an increasing rate
2. Total Product Increases at a decreasing rate

Marginal Product

Increases, comes to the highest point, gradually decreases, come to the zero, becomes negative

Average Product

Increases, comes to the highest point, gradually decreases

Reason for short run production theory

In short run when Increasing only variable inputs mixed with fixed inputs, the decrease in marginal product (MP) and average product (AP) of the variable factor after a certain point is called The Law of Diminishing Marginal Returns.

This law of diminishing marginal returns relevant to short run output is based on the following assumptions.

1. All variable inputs are homogeneous
2. Within the relevant production process technology remains constant

» Long run production process »

In long run a production firm can engage in production while changing all inputs.

Reason for Long run production theory

In long run when all inputs become variable the behavior of output can be seen in three ways.

1. Increasing returns to scale
2. Decreasing returns to scale
3. Constant returns to scale

1. Increasing returns to scale

Large proportionate increase in output more than the proportionate increase in all inputs is defined as increasing returns to scale.

Example : When increasing inputs by 20% the output increase by 30%

FEW FACTORS AFFECT INCREASING RETURNS TO SCALE.

1. Indivisibility of production factors
2. Specialization through division of labour
3. Use of machinery
4. Existence of one time cost

2. Decreasing returns to scale

Proportionate increase in output less than the proportionate increase in input is defined as decreasing returns to scale.

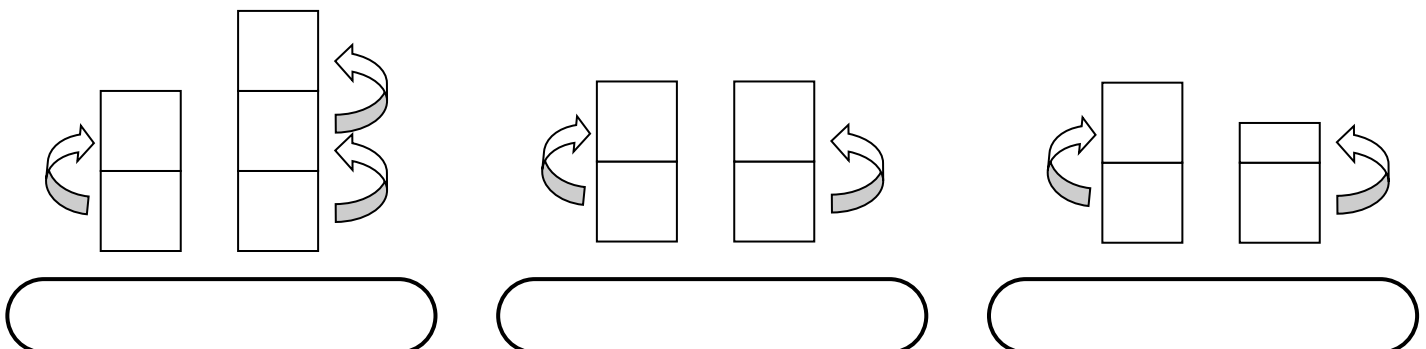
Example : When increasing inputs by 20% the output increase by 15%

- Factors affect the decrease in returns to scale :
 1. Depreciation of factors
 2. Stress
 3. Problems of coordination and management

3. Constant returns to scale

Proportionate increase in output by the same percentage of increase in input is defined as constant returns to scale

Ex : When increasing inputs by 20% the output increase by the same percentage of 20%



» COST OF PRODUCTION »

Cost of production is a phenomenon arising through a production process and it is also considered as a financial phenomenon.

» opportunity cost

Within a production process when utilizing resources for a particular use the value of the next best alternative which was able to produce using same resources is defined as opportunity cost of production process.

■ Under this concept of opportunity cost we can identify two main sub concepts.

i. Direct cost

The amount of money spent to purchase inputs from outside of the firm for production process is called direct cost.

Example :

Money spent to purchase inputs / Money spent to pay wages of workers/ Money spent for energy and electricity

ii. Indirect Cost

The opportunity cost of resources owned by the firm which used in production process is called indirect cost.

Example :

Forgone income / Forgone interest of producers / Economic depreciation / Normal profit

STAR MARK ★

An entrepreneur at present remains within a particular business activity by dedicating his labour. If he had employed his labour in another business activity he could have earned a salary from that. This salary is the amount of income forgone by this person.

Ex : Assume an entrepreneur has started a water bottling firm. If this entrepreneur worked in some other production firm and earned Rs. 60,000/- that would be the amount of income forgone by him.

Opportunity cost =	Direct cost (explicit cost)	+	Indirect Cost (Implicit cost)
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$$\text{Accounting Profit} = \text{Total Revenue} - \text{Direct Cost}$$

Example 01 :

The output of a firm equals 500 units and price of one unit equals Rs.20. The firm has spent Rs. 2000/- to purchase inputs and Rs. 1000/- to pay salaries of workers. Also the firm spent Rs. 500/- for energy and electricity. In addition Rs. 800 was spent to pay interest for borrowings. Estimate the value of profit.

01 Calculate the total revenue

02 Calculate the direct cost

03 Calculate the accounting profit

Example 02 :

The output of a firm equals 1000 units and price of one unit equals Rs.30. The firm has spent Rs. 2500/- to purchase inputs and Rs. 800/- to pay salaries of workers. Also the firm spent Rs. 200/- for energy and electricity. In addition Rs. 400 was spent to pay interest for borrowings. Estimate the value of profit.

01 Calculate the total revenue

02 Calculate the direct cost

03 Calculate the accounting profit

Example 03 :

The output of a firm equals 2500 units and price of one unit equals Rs.10. The firm has spent Rs. 500/- to purchase inputs and Rs. 3200/- to pay salaries of workers. Also the firm spent Rs. 500/- for energy and electricity. In addition Rs. 1500 was spent to pay interest for borrowings. Estimate the value of profit.

01 Calculate the total revenue

02 Calculate the direct cost

03 Calculate the accounting profit

Opportunity cost	=	Direct cost (explicit cost)	+	Indirect Cost (Implicit cost)
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Economic Cost

Economic Cost = Direct Cost + Indirect Cost

Economic Profit = Total Revenue - Economic Cost (Direct Cost + Indirect Cost)
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Ex :

output of a firm	= 500 Units	}	Direct costs
Price of one unit of output	= Rs. 20		
Wages for labour	= Rs.1000		
Purchasing of inputs	= Rs. 2000		
Payment for energy and electricity	= Rs. 500		
Interest payment for borrowing	= Rs. 800		
Forgone wage income	= Rs.500	}	Indirect Costs
Economic depreciation	= Rs.200		
Forgone interest income	= Rs.100		
Normal profit	= Rs.1000		

Total revenue = 500 x 20
= Rs. 10"000

Direct costs = 2000+ 1000+500+800
= Rs. 4300

Indirect costs = 500 + 200 + 100+ 1000
= Rs.1800

Economic profit = 10000 – (4300+1800)
= Rs. 3900

★ short run production cost. ★

Two types of inputs can be identified in a short run production firm

• Fixed factors

• Variable factors

Two types of costs can be identified in a short run production firm

• Fixed costs

• Variable cost

TOTAL FIXED COST (TFC)

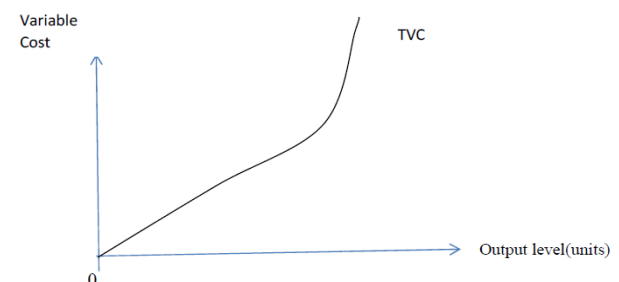
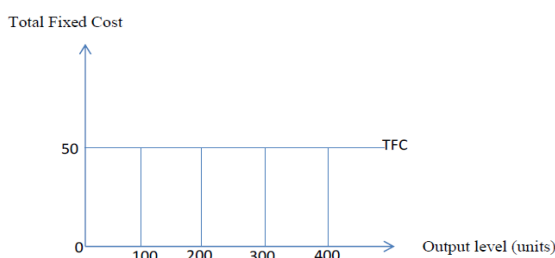
The cost bearing for fixed factors (input) related to the production process is known as total fixed cost.

» Total fixed cost does not change with the output.

There is no relationship between total fixed cost and the output level. Although the production level is zero there is a fixed cost.

EXAMPLES-

- Normal Profit ▪ Cost depreciation
- Property tax ▪ Insurance installments
- License charges ▪ Interest for capital



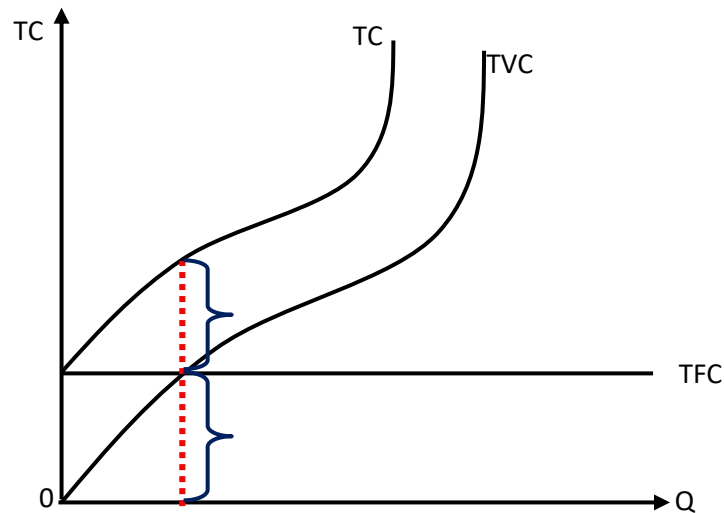
TOTAL FIXED COST



TOTAL VARIABLE COST

TOTAL COST (TC)

The total expenditure made for total variable inputs and total fixed inputs within production process is known as total cost (TC).



Using the above concepts, the average concepts related to them can be calculated.

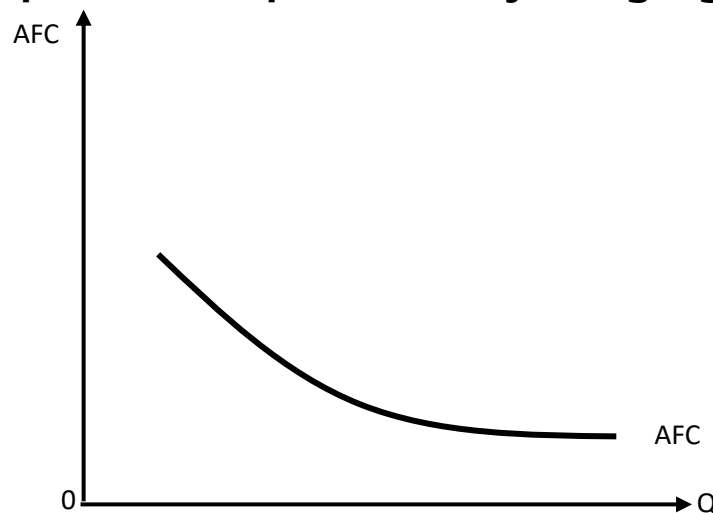
○ Average Fixed Cost (AFC)

Fixed cost per unit is known as average fixed cost.

- Average fixed cost is obtained by dividing total fixed cost from output.

$$\text{Average Fixed Cost} = \frac{\text{Total Fixed cost (TFC)}}{Q}$$

AFC curve shape can be represented by using a graph as follows



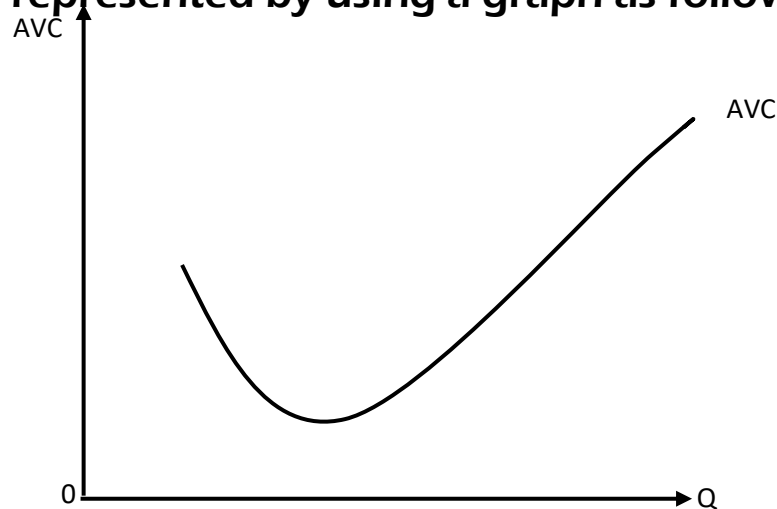
○ Average Variable Cost (AVC)

The variable cost per one unit is known as average variable cost.

- The average variable cost can be calculated by dividing the total variable cost by output level.

$$\text{Average Variable Cost} = \frac{\text{Total Variable cost (TVC)}}{Q}$$

The average variable cost curve “U” shaped. AVC curve shape can be represented by using a graph as follows

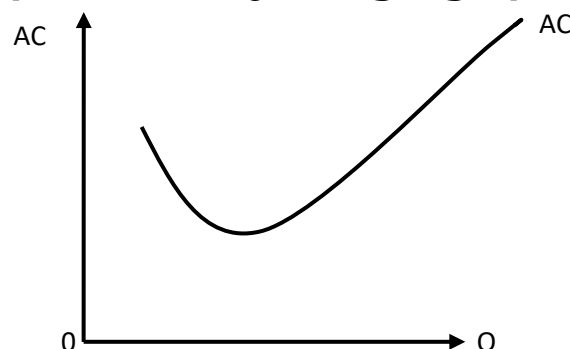


○ Average Total Cost (ATC)

The total expenditure made for total variable inputs and total fixed inputs within production process is known as total cost (TC).

$$\text{Average Cost} = \text{Average Variable cost} + \text{Average Fixed Cost}$$

The average cost curve “U” shaped. AC curve shape can be represented by using a graph as follows

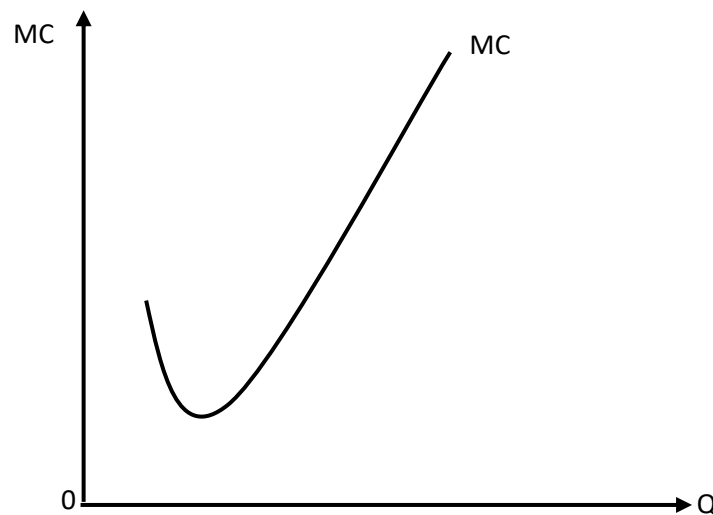


○ Marginal Cost (MC)

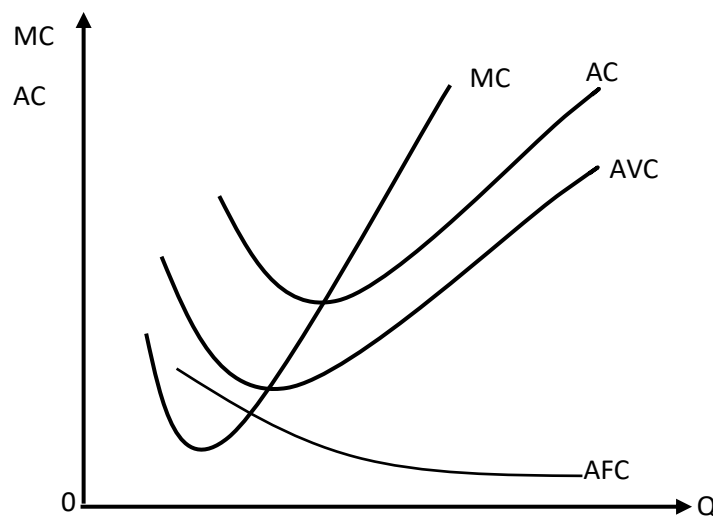
The additional to total cost, when the output is increased by one unit is called marginal cost

$$\text{Marginal Cost} = \frac{\text{Change in Total cost } (\Delta TC)}{\Delta Q}$$

Marginal cost curve is also normally “U” shaped. It can be shown by the following graph.



All these average and Marginal curves can be illustrated with the following graph.

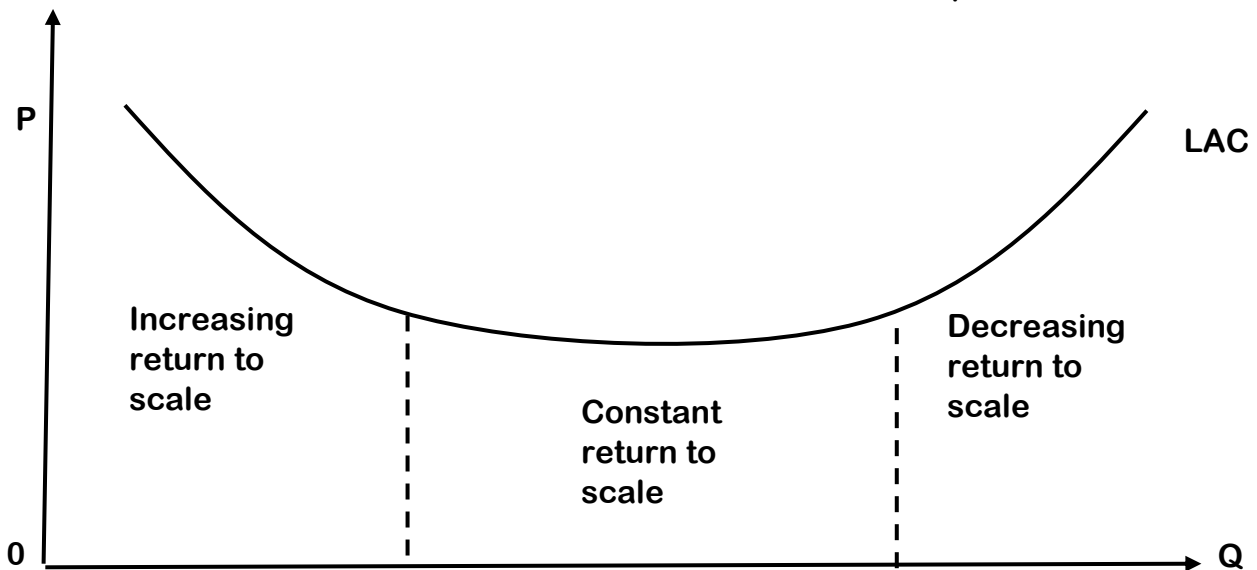


නිමැවීම ඒකක	මුළු ස්ථාවර පිරිවැය (TFC) රු	මුළු චලන පිරිවැය (TVC) රු	මුළු පිරිවැය (TC) රු	සාමාන්‍ය ස්ථාවර පිරිවැය (AFC) රු	සාමාන්‍ය චලන (AVC) රු	සාමාන්‍ය පිරිවැය (ATC) රු	ආන්තික පිරිවැය(MC) රු
0	40	0	40.00	0	0		
1	40	40.00	80.00	40.00	40.00	80.00	40
2	40	70.00					
3	40	92.00					
4	40	107.00					
5	40	117.00					
6	40	127.00					
7	40	142.00					
8	40	164.00					
9	40	194.00					
10	40	234.00					

★ Long run production cost. ★

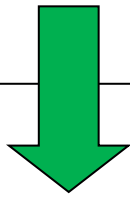
All inputs become variable inputs in the long run. There is no any barrier in the long run that affects to increase production. Due to this situation variable inputs can be seen in the long run production cost. Which is the cost for the variable inputs. According to these facts, the variable cost is equal to the total cost in the long run production process.

» **The long run average cost curve of a production firm which face economies of scale and diseconomies of scale** “U” shaped with a flat bottom



Economies of scale

Economies of scale can be defined as a financial benefit which can be obtained with decrease in the **price of inputs and increase in efficiency** of production factors due to increasing the production capacity.



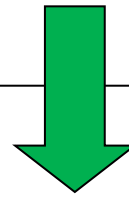
Economies of scale

Factors effecting economies of scale

1. Technology efficiencies
2. Manageral efficiencies
3. Marketing efficiencies
4. Financial efficiencies
5. Risk bearing efficiencies

Disconomies of scale

Increasing in the cost per unit (average cost) when expanding production is known as diseconomies of scale. It is the **financial loss occurred with decreasing in the efficiency** of production factors along with the expansion of production capacity.



Disconomies of scale

Factors effecting diseconomies of scale

1. Technology inefficiencies
2. Manageral inefficiencies
3. Marketing inefficiencies
4. Financial inefficiencies
5. Risk bearing inefficiencies

Average cost remains constant in the area where there is no economies of scale or diseconomies of scale. That is the area with constant returns to scale.

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○ Revenue ○



★ Total Revenue

Total revenue is the full amount of **total** sales of goods and services. It is calculated by multiplying the **total** amount of goods and services sold by the price of the goods and service.

$$\text{Total Revenue} = \text{Price} \times \text{Quantity}$$

$$TR = P \times Q$$

★ Average Revenue

Average revenue is referred to as the **revenue** that is earned per unit of output. In other words, it is the **revenue** that is obtained by the seller on selling each unit of the commodity.

$$AR = \frac{TR}{Q}$$

★ Marginal Revenue

Marginal revenue (MR) is the increase in **revenue** that results from the sale of one additional unit of output.

$$MR = \frac{\Delta TR}{\Delta Q}$$

○ Profit ●



Profit is the difference between the revenue has received from its outputs and the opportunity costs of its inputs.

$$\text{Profit} = \text{Total Revenue} - \text{Total cost}$$

★ Complete the table below using the conceptual ideas above....

Quantity	Price	Total Cost	Total Revenue	Average Revenue	Marginal revenue	Profit/loss
0	50	20				
2	50	25				
4	50	30				
6	50	38				
8	50	46				
10	50	58				
12	50	72				

★ optimum output of the firm ★

The optimal output level of profit maximization of a production firm can be analyzed in two ways.

1. Total revenue = Total cost approach

The optimal output level in terms of total revenue and total cost approach can be explained by the chart below.

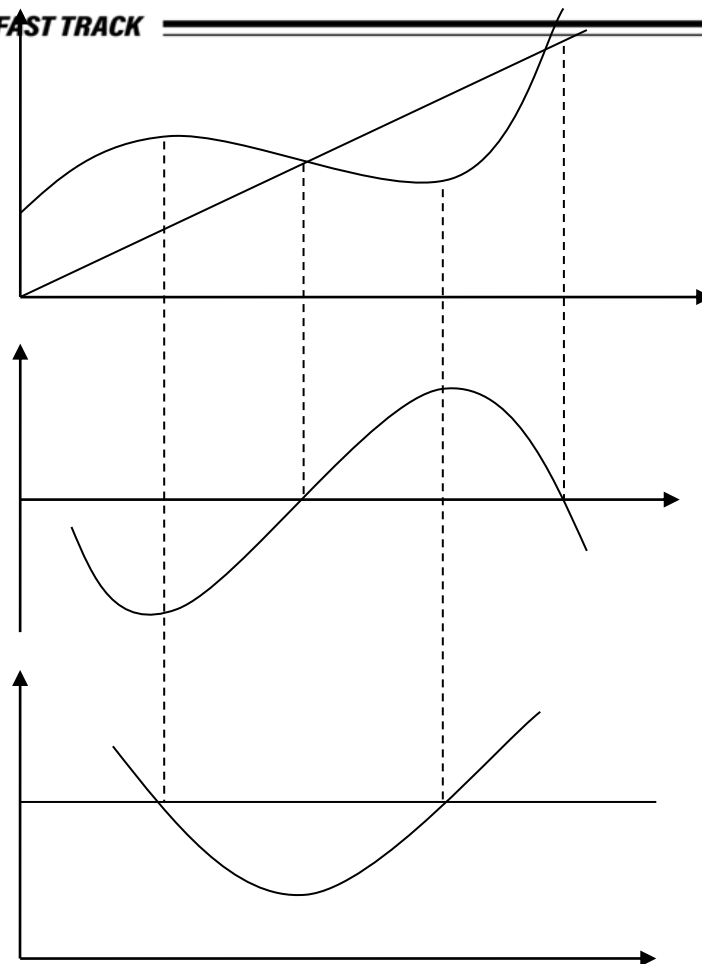
★ Accordingly, when the total revenue exceeds the total cost, the product makes a **profit**.

★ Production incurs a **loss** when the total revenue is less than the total cost.

★ Also, when the total income equals the total cost, there is no

profit or

loss

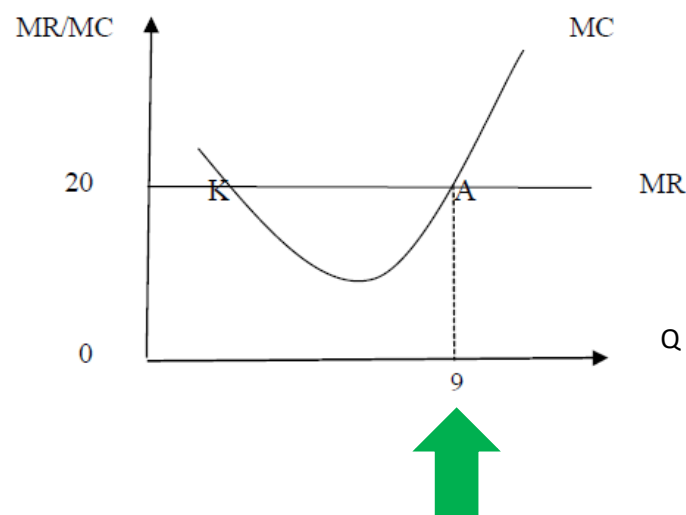


2. Marginal revenue = Marginal cost approach

The following conditions must be met for the optimal output value according to the marginal revenue and cost approach.

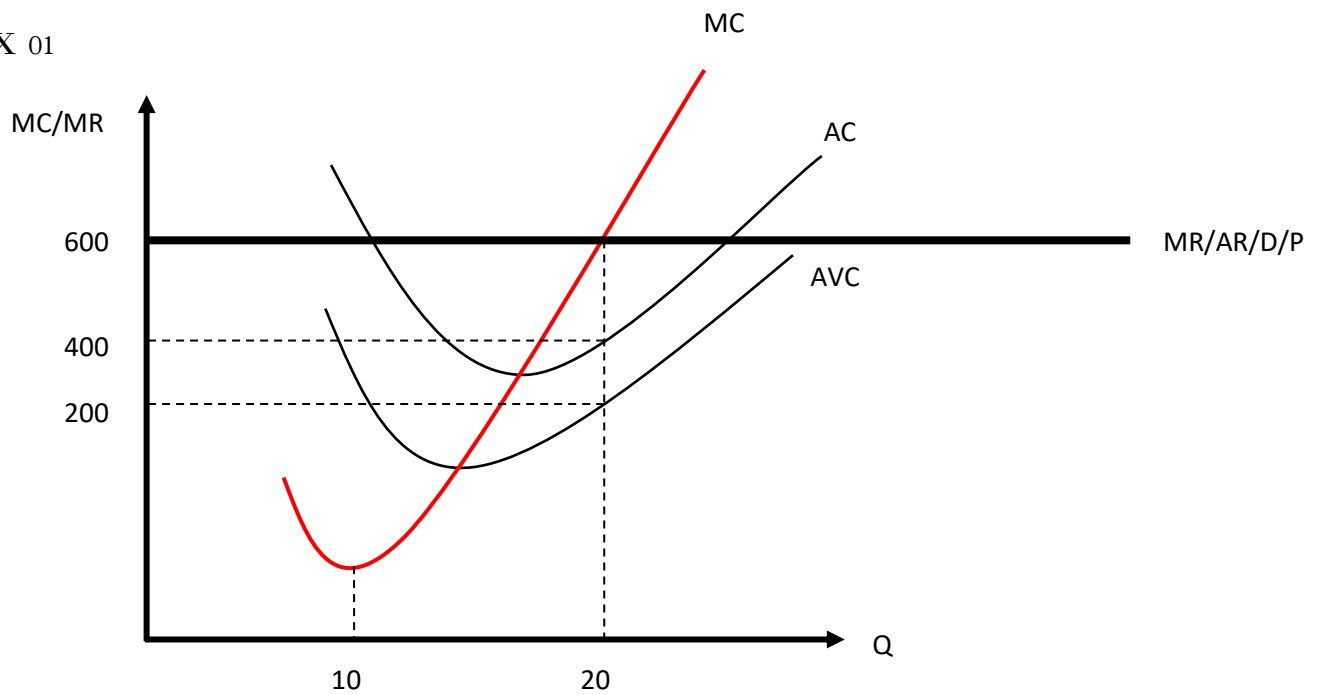
1- $MC = MR$

2- The MC curve should be increasing



Level of Maximum profit

EX 01



1 Find optimum output level

$$MR=MC$$

2 Calculate Total Revenue

$$P \times Q$$

3 Calculate Total Cost

$$AC \times Q$$

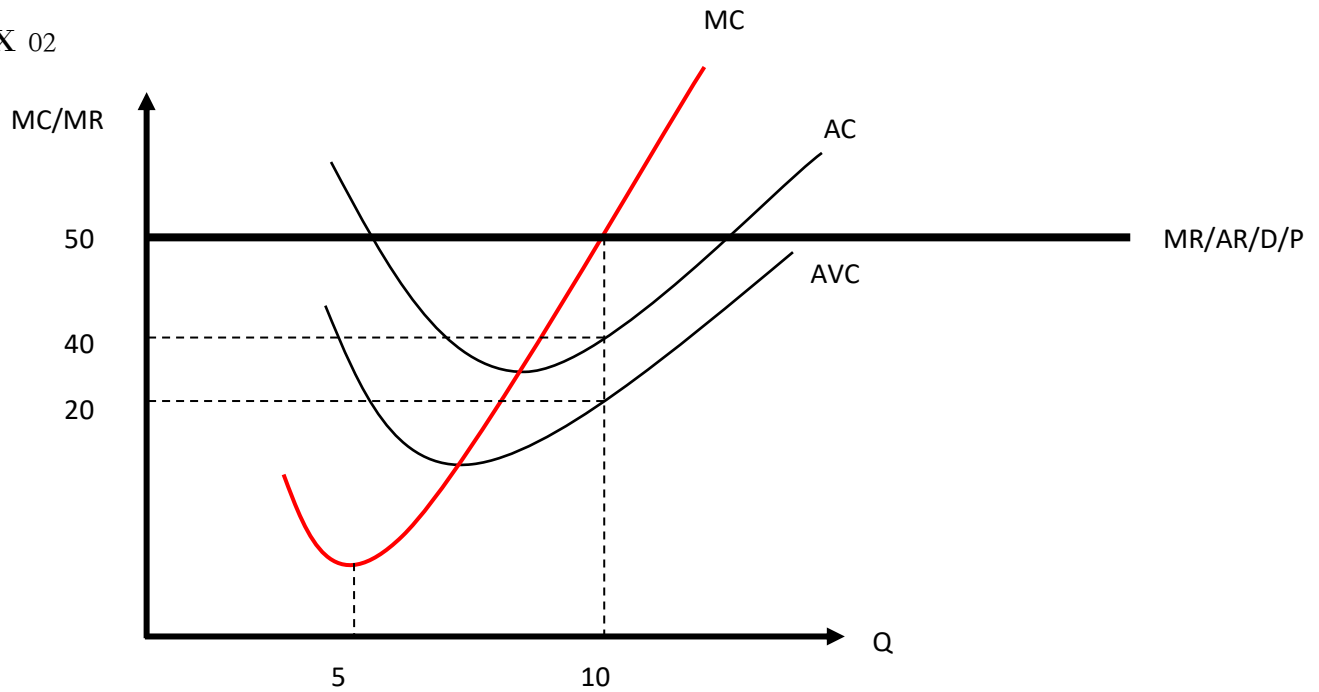
4 Calculate the profit or loss

$$TR-TC$$

5 Calculate Total Variable cost

$$AVC \times Q$$

EX 02



1 Find optimum output level

$$MR=MC$$

2 Calculate Total Revenue

$$P \times Q$$

3 Calculate Total Cost

$$AC \times Q$$

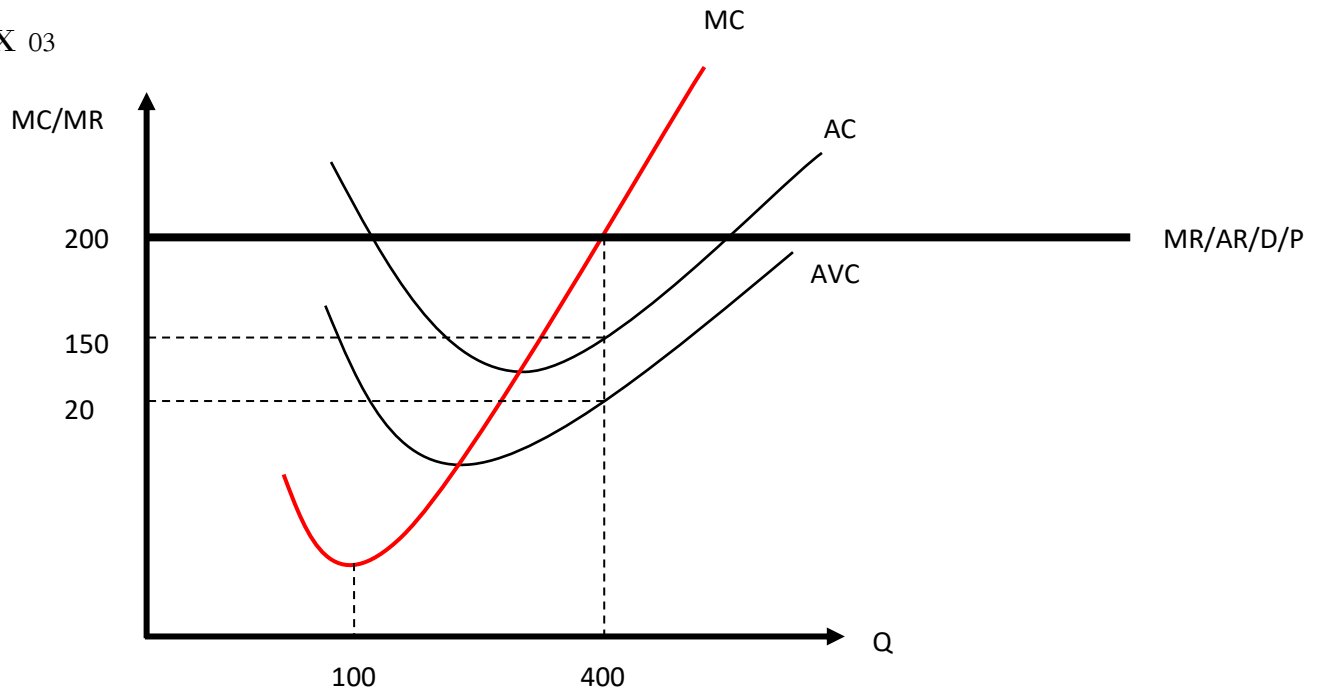
4 Calculate the profit or loss

$$TR-TC$$

5 Calculate Total Variable cost

$$AVC \times Q$$

EX 03



1 Find optimum output level

$$MR=MC$$

2 Calculate Total Revenue

$$P \times Q$$

3 Calculate Total Cost

$$AC \times Q$$

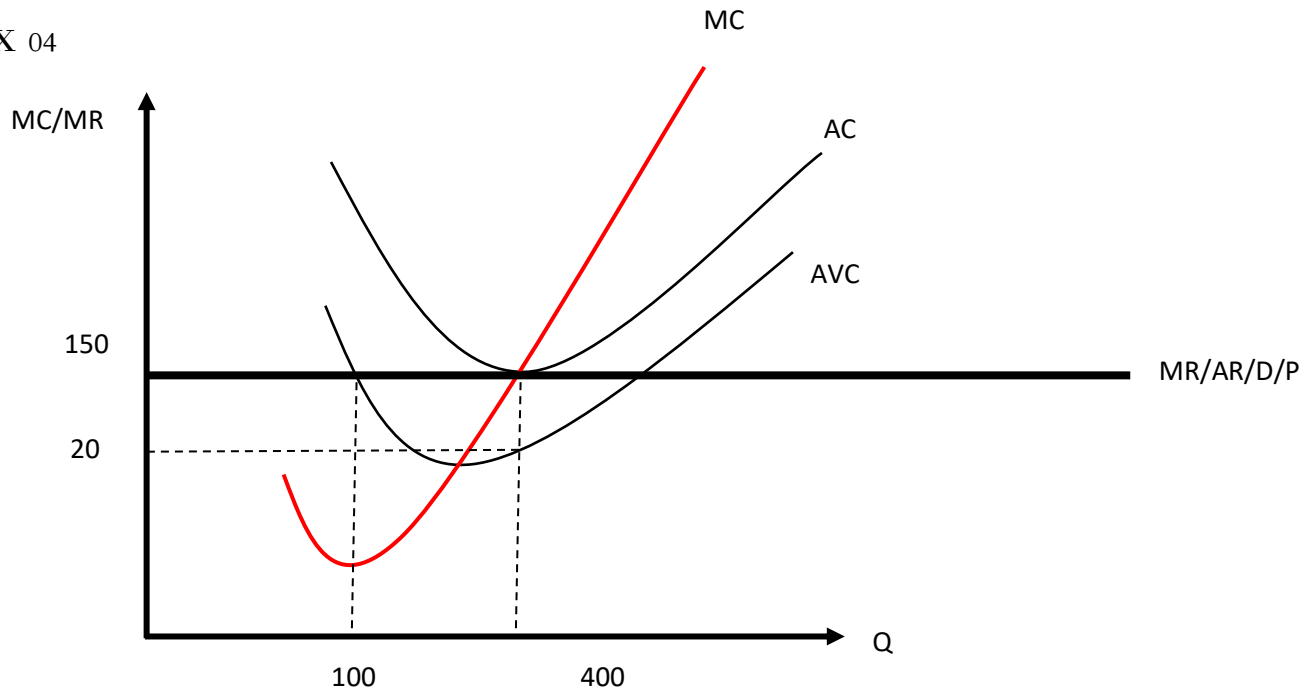
4 Calculate the profit or loss

$$TR-TC$$

5 Calculate Total Variable cost

$$AVC \times Q$$

EX 04



1 Find optimum output level

$$MR=MC$$

2 Calculate Total Revenue

$$P \times Q$$

3 Calculate Total Cost

$$AC \times Q$$

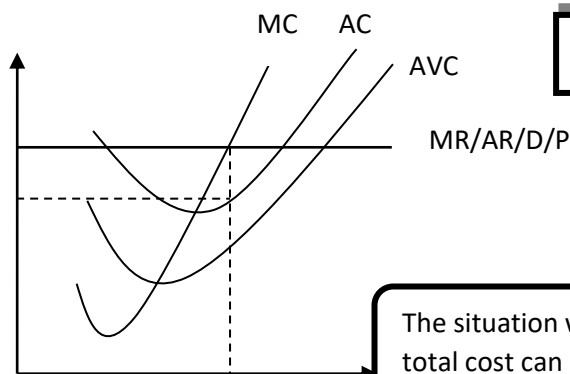
4 Calculate the profit or loss

$$TR-TC$$

5 Calculate Total Variable cost

$$AVC \times Q$$

★ Different situations in the concept of profit

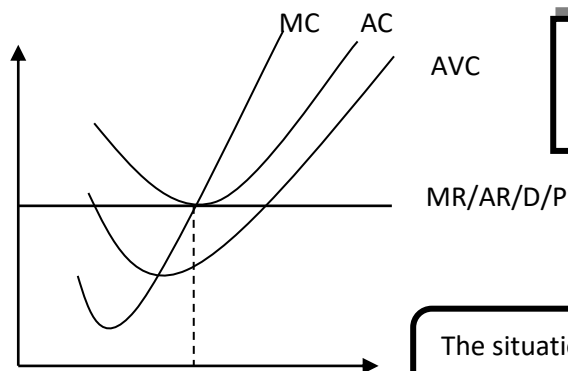


super normal profit (Abnormal)

$$AC < AR$$

$$TC < TR$$

The situation when the total revenue of the firm is higher than the total cost can be identified as super normal profit.

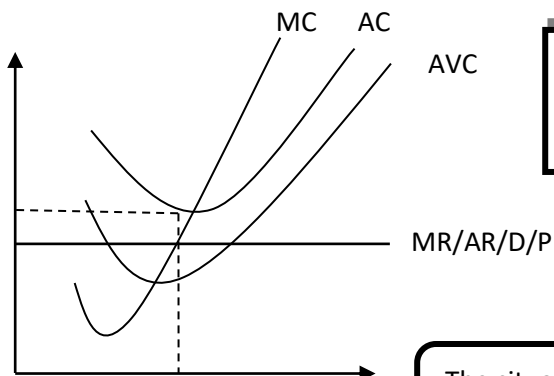


Normal profit (Zero economic profit)

$$AC = AR$$

$$TC = TR$$

The situation when the total revenue of the firm is equal to the total cost can be identified as normal profit.

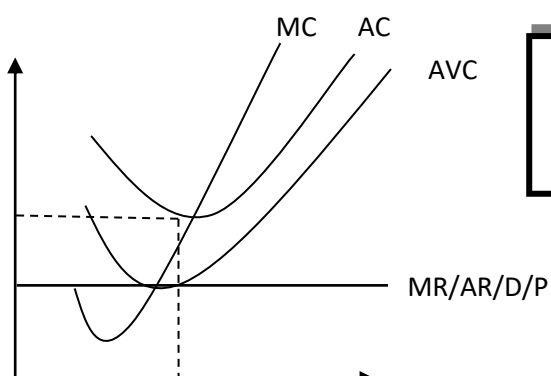


Sub normal profit (economic losses)

$$AC > AR$$

$$TC > TR$$

The situation when the total revenue of the firm is less than the total cost can be identified as economic loss.



Continues the production with losses / Closing down of the firm

$$AVC = AR$$

$$TVC = TR$$

○ Market Structures ○

The goods and services produced by a firm sell, at various market situations and to define those market situations it is used the concept of market structure.

Here you can identify different types of market shapes. It is determined by key factors such as the nature of the product being sold in the market, the number of consumer and producers.

However when studying market structures, some of their key features can be identified as follows.

★ Factors affect the various market structures :

1. Number of firms in the market

2. Homogeneity of the product

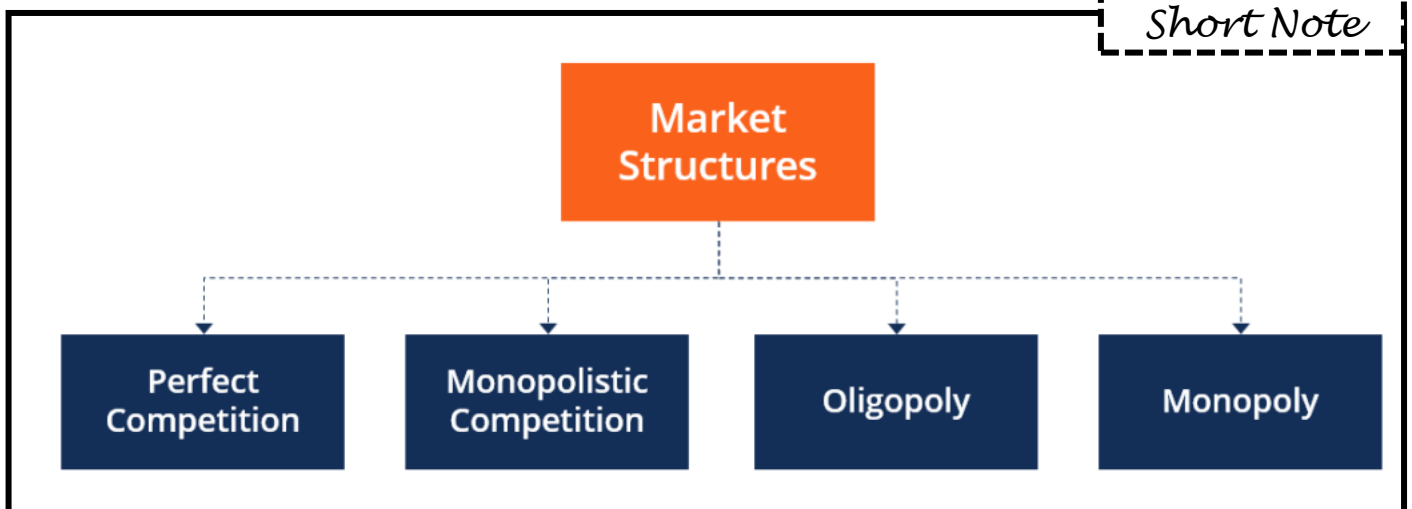
3. Ability to enter the market and exit from the market

4. The nature of the competition among the firms in the market.

Considering the above factors effect various market structures, four main types of market types can be identified as follows.

- Perfect competition.....
- Monopoly.....
- Monopolistic Competition.....
- Oligopoly.....

Short Note



★ Perfect competition ★

Market situation that has a large number of firms selling homogenous products with free entry and free exit is known as perfect competition.



○ Characteristics of perfect competition

- Homogenous products

The meaning of the homogenous product is that all production firms produce similar products and the products which are substitutable to each other. Thus it can be seen that the products of perfect competition are homogeneous with each other.

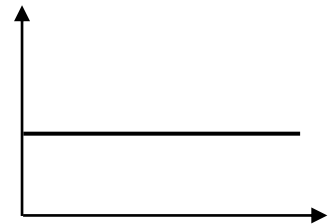
- Large number of sellers and buyers

There is no effect over the market from one individual buyer or seller as there are large number of buyers and sellers within the market.



- A Firm is a price taker

A Buyer or seller cannot change the demand or supply of the market. A perfect competition firm cannot determine the price of goods and can supply many quantity for the determined price. Because one firm cannot affect the market price and any quantity can be supplied at that market price they are known as **price takers**.



The perfectly competitive firm as a price taker because the **firms supplies a small quantity** and **produces homogenous goods**.

- Free Entrance and Exit

Free entrance is that there is no barrier for new firms to enter the market.

- There is perfect information & there are no barriers to mobility of factors



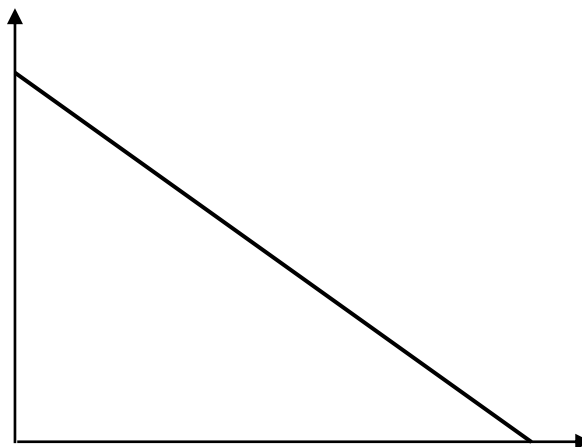
- Explain the perfect competition demand curve of a firm and the industry.
 - Market situation that has a large number of firms selling homogenous products with free entry and free exit is known as perfect competition. There are some key characteristics of this perfect competition market structure.

Characteristics:-

- * Homogenous products
- * Large number of sellers and buyers
- * A Firm is a price taker
- * Free Entrance and Exit
- * There is perfect information & there are no barriers to mobility of factors

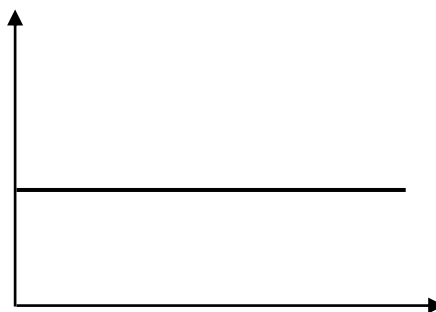
Demand curve of the perfectly competitive industry

-The total of firms which produce goods or services can be known as industry. The demand curve of the perfectly competitive industry slopes downward from left to right.

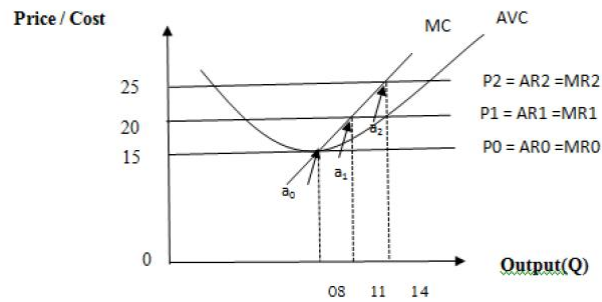


Demand curve of the perfectly competitive firm

-A unit which combines necessary resource in this way and a unit which organizes production is called a production firm. The perfectly competitive firm behaves as a price taker because the firm supplies a small quantity and produces homogenous goods. Accordingly the demand of the perfectly competitive firm can be identified as a linear line parallel to the horizontal axis.

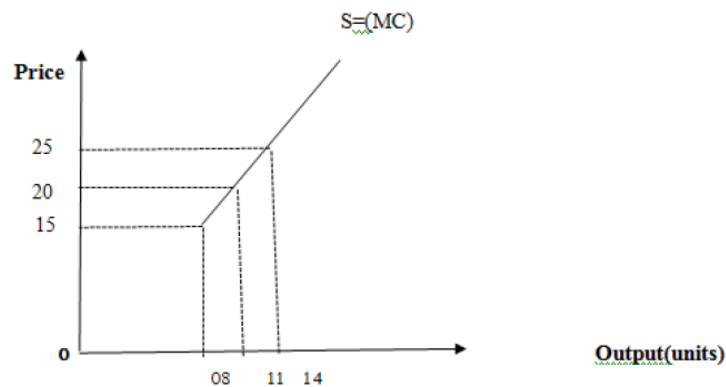


○ Perfect competition supply curve



In the above graph at point “a” price is equal to minimum variable cost and the MC curve above is the short run supply curve.

Accordingly perfectly competitive the firms’ short run supply curve can be shown in a graph as below,



★ Monopoly ★

The market which has one firm is known as the monopoly market.

A **monopoly** is a firm who is the sole seller of its product, and where there are no close substitutes. An unregulated **monopoly** has **market power** and can influence prices.

Examples: Electricity board

Srilanka Railway Department



○ Features of the monopoly market

- Unique product
- Only one firm is engaged in the production
- Barriers to entry and exit
- Imperfect information and inability to obtain information without a cost

★ Monopoly Competition ★

The market situation which consists of a large number of firms and where there are no barriers to enter the industry is known as monopolistic competition firms.

- In monopolistic competition firms have characteristics of both monopoly and perfect competition

○ characteristics of the monopolistic competition.

- Existence of large number of firms within the industry.
- Each firm produces a differentiated good.
- There are no barriers to enter and exit from the market.
- The firm's demand curve slopes downward.
- There are no barriers for factor mobility.
- Monopolistically competitive firms earn only normal profits in the long run.



Ex-

▪ Canteen



▪ Groceries



▪ Saloons



★ Oligopoly Market ★

If there is small number of firms in the market it can be defined as oligopoly market

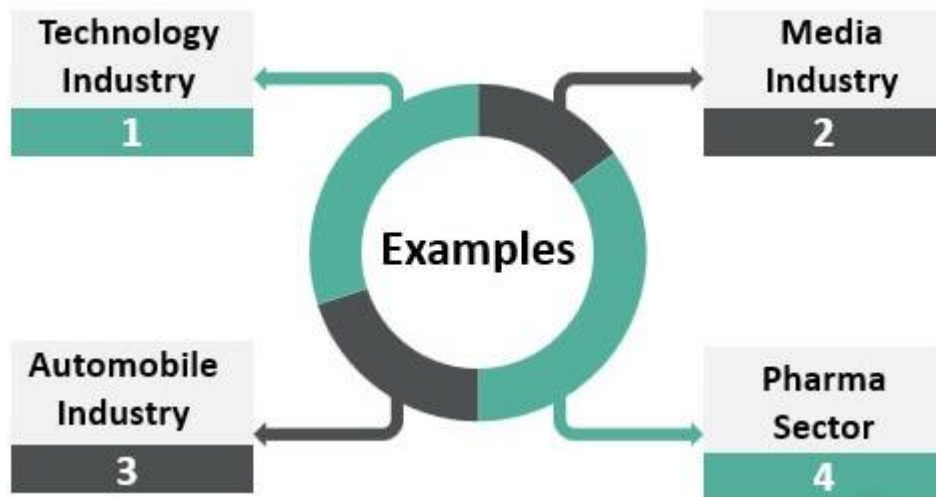
Oligopoly arises when a small number of large firms have all or most of the sales in an industry.

★ characteristics of that market

- Limited number of firm.
- Mutual understanding between firms
- Producing of either differentiated or homogeneous products
- Firm's demand curve is downward sloping and also it is relatively
- There are natural or artificial barriers which prevent entering of new firms

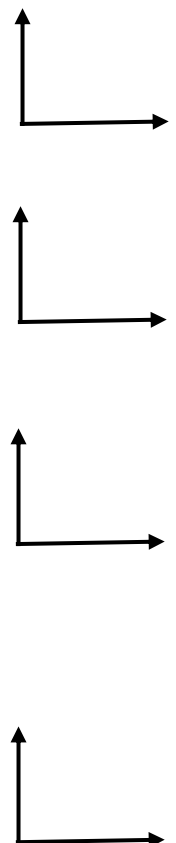


Examples -Television channels, sim card companies



★ Summary ★

Nature of the market	Number of firms	Entry to the market	Nature of product	Examples	Demand curve of the firm
Perfect competition	Very large	No barriers (Free entry)	Homogenous goods	Paddy , Corn carrot, pumpkin	Parallel to the horizontal line price taker
Monopoly	One firm	Barriers	special goods	Railway service, water supply , Electricity supply	Slope downward price maker.
Monopolistic competitive	Large	No Barriers	heterogeneous goods	canteen, grocery, saloon	Slope down ward - can affect the price - comparatively - elastic demand
Oligopoly	Few firms	Barriers	Heterogeneous or homogenous goods	News papers Television, channels commercials bank, gas, soap	Slope down ward - can effect to the price - comparatively in elastic demand



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